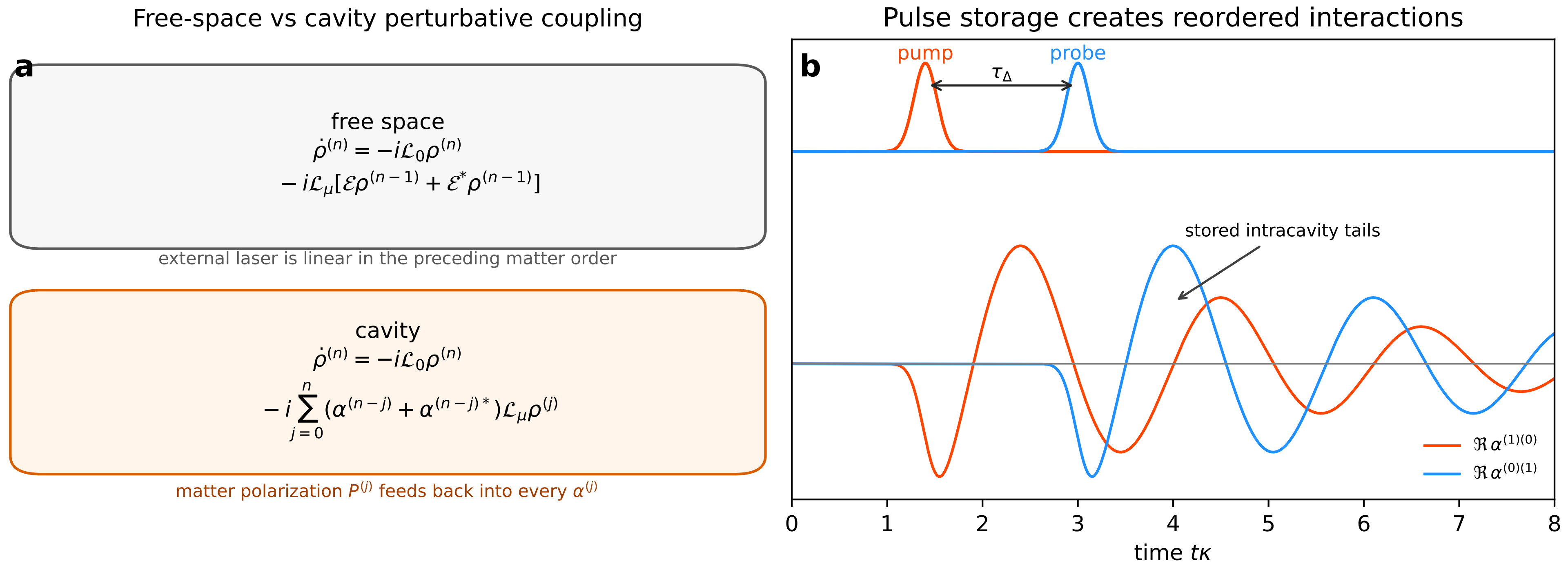
Free-space vs cavity perturbative coupling
a
free space
$\dot{\rho}^{(n)} = -i\mathcal{L}_0\rho^{(n)}$
$-i\mathcal{L}_\mu[\mathcal{E}\rho^{(n-1)} + \mathcal{E}^*\rho^{(n-1)}]$
external laser is linear in the preceding matter order
cavity
$\dot{\rho}^{(n)} = -i\mathcal{L}_0\rho^{(n)}$
$-i\sum_{j=0}^{n}(\alpha^{(n-j)} + \alpha^{(n-j)*})\mathcal{L}_\mu\rho^{(j)}$
matter polarization $P^{(j)}$ feeds back into every $\alpha^{(j)}$
Pulse storage creates reordered interactions
b
pump
probe
$\tau_\Delta$
stored intracavity tails
$\Re\,\alpha^{(1)(0)}$
$\Re\,\alpha^{(0)(1)}$
0
1
2
3
4
5
6
7
8
time $t\kappa$